

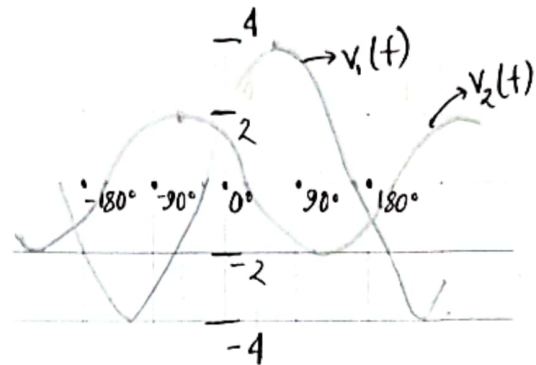
Name:

Student ID:

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Question 1 of 3 [6 Marks]
 $v_1(t) = 4 \sin(\omega t + 30^\circ)$ and $v_2(t) = 2 \cos(\omega t + 60^\circ)$.

- Which one leads by how much degree?
- Express $v_1(t)$ and $v_2(t)$ in phasor form.
- Roughly sketch $v_1(t)$ and $v_2(t)$ in the provided grid. (Label the grid, focus on phase difference)



$$\begin{aligned} a. v_1(t) &= 4 \cos(\omega t + 30^\circ - 90^\circ) \\ &= 4 \cos(\omega t - 60^\circ) \end{aligned}$$

$$v_2(t) = 2 \cos(\omega t + 60^\circ)$$

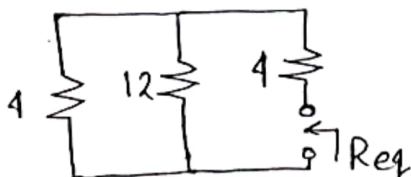
$\therefore v_2(t)$ leads $v_1(t)$ by 120° .

$$b. V_1 = 4 \angle -60^\circ$$

$$V_2 = 2 \angle 60^\circ$$

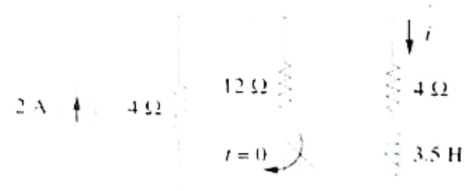
Question 2 of 3 [4 Marks]Find time constant for $t > 0$.

For $t > 0$, R_{eq} is $\rightarrow$



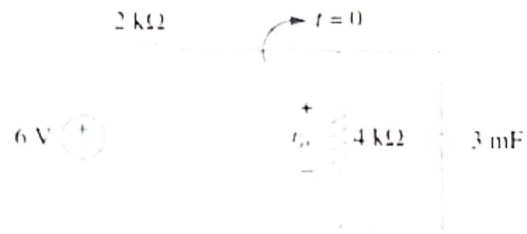
$$\begin{aligned} R_{eq} &= 4 + (4 \parallel 12) \Omega \\ &= 7 \Omega \end{aligned}$$

$$\therefore \tau = \frac{L}{R_{eq}} = \frac{3.5}{7} = 0.5 \text{ sec}$$

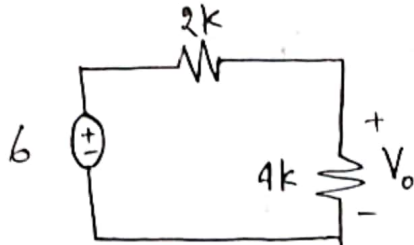


Question 3 of 3 [10 Marks]

- Find v_o for $t < 0$.
- Find time constant for $t > 0$.
- Find the expression of capacitor voltage for $t > 0$.
- What is the capacitor voltage at $t = 2\text{ms}$?



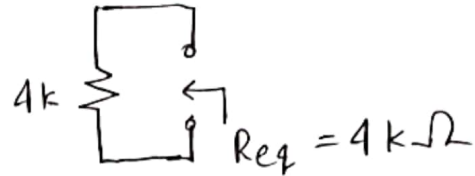
a. For $t < 0$,



$$V_o = \frac{4k}{2k+4k} \times 6$$

$$= 4V.$$

b. For $t > 0$, R_{eq} is $\rightarrow$



$$\therefore \tau = R_{eq} C = 4k \times 3m$$

$$= 12 \text{ sec}$$

c. For source-free RC ckt, $v_c(t) = V_o e^{-t/\tau}$

$$= 4 e^{-t/12} \text{ V.}$$

d. $v_c(t = 2 \times 10^{-3}) = 4 e^{-2 \times 10^{-3}/12} \text{ V}$

$$= 3.999 \text{ V.}$$

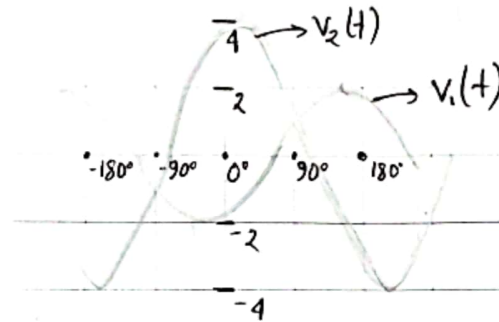
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Question 1 of 3 [6 Marks]
 $v_1(t) = 2 \sin(\omega t - 60^\circ)$ and $v_2(t) = 4 \cos(\omega t - 30^\circ)$.

- Which one leads by how much degree?
- Express $v_1(t)$ and $v_2(t)$ in phasor form.
- Roughly sketch $v_1(t)$ and $v_2(t)$ in the provided grid. (Label the grid, focus on phase difference)



$$\begin{aligned} a. \quad v_1(t) &= 2 \cos(\omega t - 60^\circ - 90^\circ) \\ &= 2 \cos(\omega t - 150^\circ) \end{aligned}$$

$$v_2(t) = 4 \cos(\omega t - 30^\circ)$$

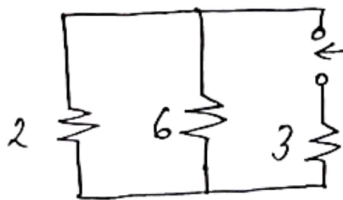
$\therefore v_2(t)$ leads $v_1(t)$ by 120° .

$$b. \quad V_1 = 2 \angle -150^\circ$$

$$V_2 = 4 \angle -30^\circ$$

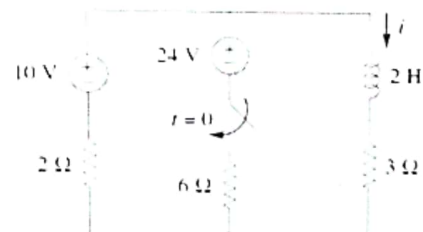
Question 2 of 3 [4 Marks]Find time constant for $t > 0$.

For $t > 0$, R_{eq} is $\rightarrow$



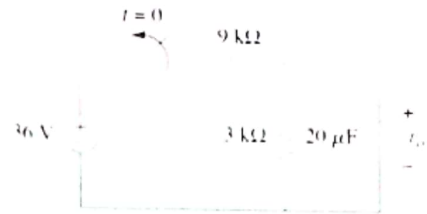
$$\begin{aligned} R_{eq} &= 3 + (2 \parallel 6) \\ &= 4.5 \Omega \end{aligned}$$

$$\therefore \tau = \frac{L}{R_{eq}} = \frac{2}{4.5} = 0.444 \text{ sec.}$$

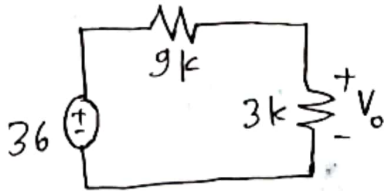


Question 3 of 3 [10 Marks]

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- Find time constant for $t > 0$.
- Find the expression of capacitor voltage for $t > 0$.
- What is the capacitor voltage at $t = 2\text{ms}$?



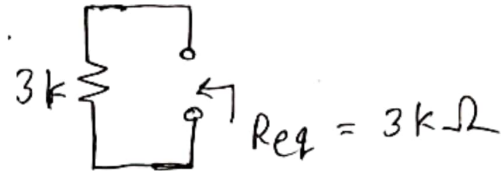
a. For $t < 0$,



$$v_o = \frac{3k}{3k + 9k} \times 36$$

$$= 9V.$$

b. For $t > 0$, R_{eq} is $\rightarrow$



$$\therefore \tau = R_{eq} C = 3k \times 20\mu = 60\text{ms}.$$

c) For source-free RC ckt, $v_c(t) = v_o e^{-t/\tau}$

$$= 9 e^{-t/(60 \times 10^{-3})} V.$$

d) $v_c(t = 2 \times 10^{-3}) = 9 e^{(-2 \times 10^{-3})/(60 \times 10^{-3})} V$

$$= 8.705V.$$

